* Find T^{-1}.

$$T(1,0) = (2, 1)$$

$$T(0,1) = (3, -1)$$

$$[T]_S = \begin{bmatrix} 2 & 3 \\ 1 & -1 \end{bmatrix}$$

$$T^{-1} = \frac{-1}{5}\begin{bmatrix} -1 & -3 \\ -1 & -2 \end{bmatrix}$$

$$T^{-1}(x,y) = \left(\frac{1}{5}x + \frac{3}{5}y, \frac{1}{5}x + \frac{2}{5}y\right)$$

* $[P]_{SS'}[v]_{S'} = [v]_S$ — prove this

$$[P^{-1}(v)]_S = [v]_{S'}$$

$$Q_{S'S}[v]_S = [v]_{S'}$$

Result:-

Let P be the change of basis matrix from S to a basis S' in a vectorspace V, then for any linear operator T on V,

$$[T]_{S'} = P^{-1}[T]_S P$$

* Two matrices represent the same linear operator iff matrices are similar.

P Kalika Maths